

Topic: Selection Operation

Q. Explain the Selection operation in Relational Algebra. Write its syntax and provide examples demonstrating how it filters tuples.

 **Definition to Remember**

The **Selection Operation** (denoted by σ) is a **unary operation** in relational algebra that filters and returns specific **rows (tuples)** from a relation that satisfy a given logical condition (predicate). It is equivalent to the **WHERE** clause in SQL.

Selection Operation (σ)

- Operates on a **single table** (unary).
- Filters **horizontally** — returns same columns, but only matching rows.
- Output automatically eliminates duplicate rows (set behavior).

2. Syntax of Selection

$\sigma_{\{ \}}(\text{Relation_Name})$

- Predicate:** Uses comparison operators ($=, \neq, <, \leq, >, \geq$) and logical operators (AND AND, OR OR, NOT NOT).

3. Examples

Input Relation: **EMPLOYEE**

Emp_ID	Name	Department	Salary
1	Alice	IT	50000
2	Bob	HR	40000
3	Charlie	IT	60000
4	David	Sales	45000

Example 1: Simple Condition

Query: Find all IT department employees.

$\sigma_{\{\text{Department} = \text{'IT'}\}}(\text{EMPLOYEE})$

Result: Returns rows for Alice (ID 1) and Charlie (ID 3).

Example 2: AND Condition (AND)

Query: IT employees with salary > 55000.

$\sigma_{\{\text{Department} = \text{'IT'} \text{ AND } \text{Salary} > 55000\}}(\text{EMPLOYEE})$

Result: Returns only Charlie (IT, 60000).

Example 3: OR Condition (OR)

Query: Employees in HR or Sales.

$\sigma_{\{\text{Department} = \text{'HR'} \text{ OR } \text{Department} = \text{'Sales'}\}}(\text{EMPLOYEE})$

Result: Returns Bob (HR) and David (Sales).

★ Must-Write Points (for 10 marks)

1. Selection is a **unary operation** — operates on one relation.
2. Denoted by σ (**Sigma**); equivalent to SQL's **WHERE** clause.
3. Filters **rows (tuples)** based on a logical predicate — horizontal filtering.
4. Output has the same columns as input but only matching rows.
5. Predicates use comparison operators ($=$, $\neq$, $<$, $>$) and logical operators ($\wedge$, $\vee$, $\neg$).
6. Can be **nested**: apply selection to the result of another operation.
7. Automatically eliminates duplicate rows (set semantics).

⚡ Quick Recall

σ (Sigma) $\rightarrow$ Unary $\rightarrow$ Horizontal Filtering (rows) $\rightarrow$ Predicate ($=$, $\neq$, $\wedge$, $\vee$) $\rightarrow$ Equivalent to SQL WHERE $\rightarrow$ Can be nested